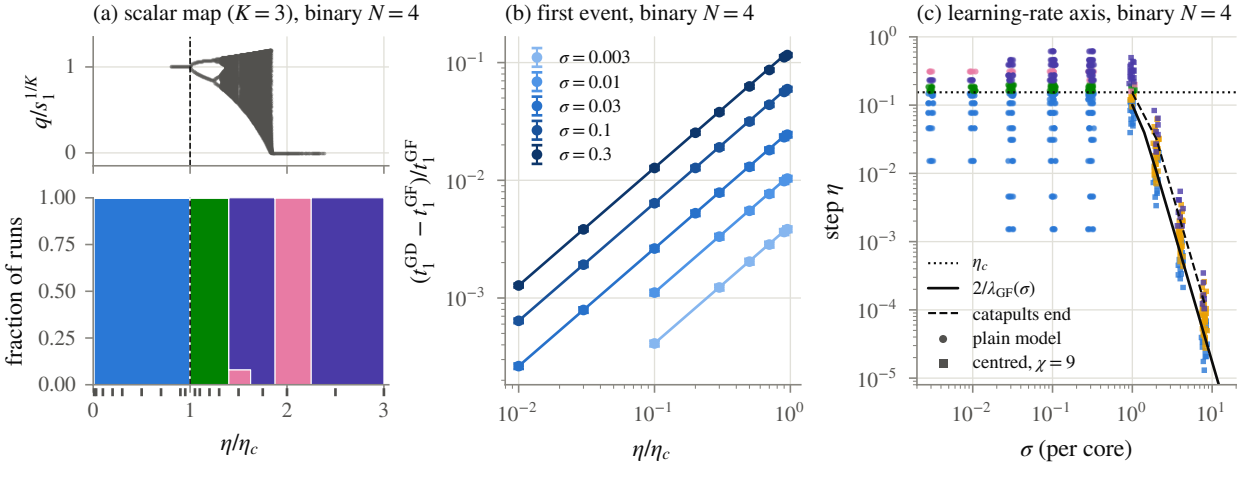
(a) scalar map ($K = 3$), binary $N = 4$
$q/s_1^{1/K}$
1
0
fraction of runs
1.00
0.75
0.50
0.25
0.00
0
1
2
3
η/η_c
(b) first event, binary $N = 4$
$(t_1^{\mathrm{GD}} - t_1^{\mathrm{GF}})/t_1^{\mathrm{GF}}$
$\sigma = 0.003$
$\sigma = 0.01$
$\sigma = 0.03$
$\sigma = 0.1$
$\sigma = 0.3$
10^{-1}
10^{-2}
10^{-3}
10^{-2}
10^{-1}
10^{0}
η/η_c
(c) learning-rate axis, binary $N = 4$
step η
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
η_c
$2/\lambda_{\mathrm{GF}}(\sigma)$
catapults end
plain model
centred, $\chi = 9$
10^{-2}
10^{-1}
10^{0}
10^{1}
σ (per core)
GF-like
catapult
bounded oscillation
thrown back to the origin
diverged